Abstract linear algebra homework7

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Problem 1

Statement. Let $A \in M_n(\mathbb{C})$ be invertible, and assume that A^k is diagonalizable over $\mathbb{C}$ for some $k \geq 2$. Prove that A is diagonalizable over $\mathbb{C}$.

Proof. Since A^k is diagonalizable, its minimal polynomial factors as

$$m_{A^k}(x) = \prod_{i=1}^{m}(x - \lambda_i),$$

where $\lambda_1, \ldots, \lambda_m \in \mathbb{C}$ are distinct. Since A is invertible, $\det(A) \neq 0$, so $\det(A^k) = \det(A)^k \neq 0$, and hence A^k is invertible. In particular 0 is not an eigenvalue of A^k, so 0 is not a root of m_{A^k}; that is, $\lambda_i \neq 0$ for all i.

Since $\mathbb{C}$ is algebraically closed, each factor $x^k - \lambda_i$ splits into linear factors over $\mathbb{C}$:

$$x^k - \lambda_i = \prod_{j=0}^{k-1}(x - \mu_{i,j}),$$

where $\mu_{i,0}, \ldots, \mu_{i,k-1}$ are the k (distinct, since $\lambda_i \neq 0$) k-th roots of λ_i in $\mathbb{C}$.

Claim. *The mk numbers $\{\mu_{i,j} : 1 \leq i \leq m,\ 0 \leq j \leq k-1\}$ are pairwise distinct.*

Proof of claim. Within a fixed i, the roots $\mu_{i,0}, \ldots, \mu_{i,k-1}$ are distinct by construction (the k-th roots of a nonzero complex number are k distinct points). Across different indices, suppose $\mu_{i,j} = \mu_{i',j'}$ for some $i \neq i'$. Since $x \mapsto x^k$ is a well-defined (single-valued) function on $\mathbb{C}$, this common value μ satisfies both $\mu^k = \lambda_i$ and $\mu^k = \lambda_{i'}$, forcing $\lambda_i = \lambda_{i'}$ — contradicting that the λ_i are distinct. So no such coincidence can occur, and all mk roots are distinct. □

By definition of the minimal polynomial, $m_{A^k}(A^k) = 0$. Substituting $x = A$ into the factorization above (legitimate since evaluation at a fixed matrix A is an algebra homomorphism on polynomials — all powers of A commute with each other),

$$0 = m_{A^k}(A^k) = \prod_{i=1}^{m}\left(A^k - \lambda_i I\right) = \prod_{i=1}^{m}\prod_{j=0}^{k-1}(A - \mu_{i,j}I).$$

So A is annihilated by the polynomial $\prod_{i,j}(x - \mu_{i,j})$, which is a product of mk *pairwise distinct* linear factors (by the Claim). Hence the minimal polynomial $m_A(x)$ of A divides this product, so $m_A(x)$ itself is a product of distinct linear factors. Since a matrix over $\mathbb{C}$ is diagonalizable if and only if its minimal polynomial splits into distinct linear factors, we conclude that A is diagonalizable over $\mathbb{C}$. □

Problem 2

(a) $V = \mathbb{R}^2$, $T(x, y) = (-y, x)$

The matrix of T in the standard basis is $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Proof. First compute T^2:

$$T^2(x, y) = T(-y, x) = (-x, -y) = -(x, y).$$

So $T^2 = -I$, i.e. $T^2 + I = 0$. Thus $q(x) = x^2 + 1$ annihilates T, and the minimal polynomial $m_T(x)$ divides $q(x)$.

Since $q(x) = x^2 + 1$ is irreducible over $\mathbb{R}$ (it has no real roots), its only monic divisors in $\mathbb{R}[x]$ are 1 and $x^2 + 1$ itself. The constant polynomial 1 does not annihilate T (that would require $T = 0$, which is false). So $m_T(x)$ must equal $q(x)$, provided we also rule out a possible degree-1 factor. But since q is irreducible of degree 2, its only monic factors are 1 and q itself, so this is automatic.

Alternatively, directly: if $m_T(x) = x - c$ for some $c \in \mathbb{R}$, then $T = cI$ would have to be a scalar multiple of the identity. But $T(1, 0) = (0, 1) \neq c(1, 0)$ for any real c unless $c = 0$, and $c = 0$ fails too since $T \neq 0$. So T has no real eigenvalues, ruling out any linear annihilating polynomial.

Therefore

$$m_T(x) = x^2 + 1.$$

□

(b) $V = \mathbb{R}^2$, $T(x, y) = (y, x)$

Proof. Compute T^2:

$$T^2(x, y) = T(y, x) = (x, y).$$

So $T^2 = I$, i.e. $(T - I)(T + I) = T^2 - I = 0$. Thus $q(x) = x^2 - 1 = (x - 1)(x + 1)$ annihilates T, so $m_T(x) \mid (x - 1)(x + 1)$.

We rule out the linear factors individually. If $m_T(x) = x - 1$, then $T = I$; but $T(1, 0) = (0, 1) \neq (1, 0)$, so $T \neq I$. If $m_T(x) = x + 1$, then $T = -I$; but $T(1, 0) = (0, 1) \neq (-1, 0)$, so $T \neq -I$. Hence neither linear polynomial annihilates T, and since $(x - 1)$ and $(x + 1)$ are the only monic proper divisors of $(x - 1)(x + 1)$, we conclude

$$m_T(x) = (x - 1)(x + 1) = x^2 - 1.$$

□

(c) $V = \mathbb{R}^4$, $T(x, y, z, w) = (-y, x, w, z)$

Proof. Observe that T preserves the decomposition $V = U_1 \oplus U_2$, where $U_1 = \{(x, y, 0, 0)\}$ and $U_2 = \{(0, 0, z, w)\}$: indeed T maps U_1 into U_1 via $(x, y, 0, 0) \mapsto (-y, x, 0, 0)$, which is exactly the rotation operator R from part (a), and T maps U_2 into U_2 via $(0, 0, z, w) \mapsto (0, 0, w, z)$, which is exactly the swap operator S from part (b). So $T = R \oplus S$ with respect to this splitting.

An annihilating polynomial. From part (a), $R^2 = -I_{U_1}$, so $R^4 = I_{U_1}$. From part (b), $S^2 = I_{U_2}$, so certainly $S^4 = I_{U_2}$. Hence $T^4 = R^4 \oplus S^4 = I_{U_1} \oplus I_{U_2} = I_V$. (One can verify this directly too: $T^2(x, y, z, w) = T(-y, x, w, z) = (-x, -y, z, w)$, and $T^4(x, y, z, w) = T^2(-x, -y, z, w) = (x, y, z, w)$.) So $q(x) = x^4 - 1$ annihilates T, and $m_T(x) \mid x^4 - 1 = (x - 1)(x + 1)(x^2 + 1)$.

Minimality. Suppose $g(x)$ is any polynomial with $g(T) = 0$. Restricting to the invariant subspace U_1, $g(T)|_{U_1} = g(R) = 0$, so g is annihilated by R, and thus the minimal polynomial of R (which is $x^2 + 1$, by part (a)) divides g. Similarly, restricting to U_2, $g(S) = 0$, so the minimal polynomial of S (which is $x^2 - 1$, by part (b)) divides g.

Thus g must be a common multiple of $x^2 + 1$ and $x^2 - 1$. These two polynomials are coprime in $\mathbb{R}[x]$: $x^2 + 1$ is irreducible over $\mathbb{R}$ and does not divide $x^2 - 1$ (their root sets $\{\pm i\}$ and $\{\pm 1\}$ are disjoint), so $\gcd(x^2 + 1, x^2 - 1) = 1$. Hence any common multiple of $x^2 + 1$ and $x^2 - 1$ must be a multiple of their product,

$$\operatorname{lcm}(x^2 + 1,\, x^2 - 1) = (x^2 + 1)(x^2 - 1) = x^4 - 1.$$

In particular $\deg g \geq 4$, i.e. no polynomial of degree less than 4 annihilates T.

Since $q(x) = x^4 - 1$ annihilates T and has degree 4, and we've shown any annihilating polynomial has degree ≥ 4, we conclude

$$m_T(x) = x^4 - 1.$$

□

Problem 13

Proof. Since ST is nilpotent, there exists $n \in \mathbb{Z}^+$ such that $(ST)^n = 0$.

Consider $(TS)^{n+1}$. Writing out the product of $n + 1$ copies of TS and regrouping,

$$(TS)^{n+1} = \underbrace{(TS)(TS)\cdots(TS)}_{n+1 \text{ factors}} = T\,\underbrace{(ST)(ST)\cdots(ST)}_{n \text{ factors}}\,S = T(ST)^n S.$$

Since $(ST)^n = 0$, this gives

$$(TS)^{n+1} = T \cdot 0 \cdot S = 0.$$

Hence TS is nilpotent (with $(TS)^{n+1} = 0$). □

Problem 14

Proof. We argue by contradiction. Suppose T is diagonalizable. Then V has a basis $v_1, \ldots, v_d$ consisting of eigenvectors of T, say $Tv_i = \lambda_i v_i$ with $\lambda_i \in F$.

Since T is nilpotent, there exists $n \geq 1$ with $T^n = 0$. For each basis vector v_i,

$$0 = T^n v_i = \lambda_i^n v_i.$$

Since $v_i \neq 0$ (it is a basis vector), we must have $\lambda_i^n = 0$, and since we are working over a field, this forces $\lambda_i = 0$.

This holds for every $i = 1, \ldots, d$, so *every* eigenvalue of T is 0. But then, in the eigenbasis $v_1, \ldots, v_d$, the matrix of T is the zero matrix, which means $Tv_i = 0$ for every basis vector v_i, and hence $T = 0$ on all of V (since the v_i span V).

This contradicts the hypothesis $T \neq 0$. Therefore T is not diagonalizable. □

Check-In

Everything is absolutely great. I gained a lot of new understandings of the definitions and theorems I have learned, also the motivation seemed to be more clearly! Nice job Akash!!!!